Fast & Curious

Ramp height, displacement, and inferred energy of motion · Friday PRACTICE

Name: _____ **Date:** _____

THE DATA FILE

A student recorded two trials for each ramp height. Complete the table.

Ramp height (books)	Trial 1 push (cm)	Trial 2 push (cm)	Average push (cm)
1 book	3	5	
2 books	7	9	
3 books	11	13	
4 books	15	17	

MEASURED VS. INFERRED

Ramp height — MEASURED / INFERRED — why? _____

Cup push (displacement) — MEASURED / INFERRED — why? _____

Relative speed of the marble — MEASURED / INFERRED — why? _____

Relative energy of motion — MEASURED / INFERRED — why? _____

MAKE A CLAIM

The SAME marble rolled from a HIGHER ramp made the cup move _____ (more / less).
A faster roll causes a _____ (greater / smaller) change, so it has _____ (more / less) energy of motion.

TEACHER KEY

Averages: 1 book = 4 · 2 books = 8 · 3 books = 12 · 4 books = 16

Measured: ramp height, cup push (read with a ruler/tape).

Inferred: relative speed (from ramp height) and relative energy of motion (from the cup push — we infer energy from the change we see).

Claim: more / greater / more. (Same object, faster = more energy of motion = greater change.)

No speed is directly measured and no m/s value is invented.